Functional Analysis

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1 Jan 9

Assume knowledge of 403 and 531 basically. Dunford & Schwartz. Recall the definition of a *normed vector space*. It induces a metric and hence a topology. A metric space is a Banach space if it is complete. It's often that if you have infinite-dim vector spaces, you have a lot of topologies to choose from, each of which have different roles. We will find out how to utilize each of them. Some examples are $\mathbb{R}^n$ with the p-norm, ℓ^p such that $\left(\sum_{j=1}^{\infty} |x_j|^p\right) < \infty$, or $L^p(\mathbb{R}^n)$. We will begin with the simplest topology.

Definition 1.1 (Bounded Linear Operators)

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be 2 normed spaces. Then, a linear map $A : X \to Y$ is called **bounded** if there exists a $c \geq 0$, s.t. $\forall v \in X$,

$$\|Av\|_Y \leq c\|v\|_X \tag{1}$$

The smallest such c is denoted $\|A\| = \|A\|_{\text{sup}} = \|A\|_\infty$, i.e. the spectral norm. i.e.

$$\|A\| \coloneqq \sup_{w \in X \setminus \{0\}} \frac{\|Aw\|_Y}{\|w\|_X} \tag{2}$$

Note that due to linearity, we can take the sup over the unit sphere as well.

Set

$$L(X, Y) \coloneqq \{A : X \to Y \coloneqq A \text{ linear, bounded}\} \tag{3}$$

which is a normed vector space. The topology is called the *uniform operator topology*.

Theorem 1.1 (1.2.2.)

Let X, Y be two normed vector spaces. Let $A : X \to Y$ be linear. TFAE.

1. A is bounded.
2. A is continuous.
3. A is continuous at 0 (or any point).

Proof. Listed.

1. $(1 \to 2)$. The one-liner is that A is Lipschitz, which is continuous, since

$$\|Au - Av\|_Y = \|A(u - v)\|_Y \leq \|A\|\|u - v\| \tag{4}$$

2. $(2 \to 3)$. Clear.
3. $(3 \to 1)$. For every $\epsilon > 0$, $\exists \delta_\epsilon > 0$ s.t.

$$A^{-1}\big(B_Y(0, \epsilon)\big) \supset B_X(0, \delta_\epsilon) \tag{5}$$

where A^{-1} indicates the preimage. In other words, $|u|_X < \delta_\epsilon \implies |Au|_Y < \epsilon$. Let $w \in X \setminus \{0\}$. Then,

$$w = |w| \cdot \frac{2}{\delta_\epsilon} \frac{w}{|w|} \frac{\delta_\epsilon}{2} \tag{6}$$

So by linearity, we have

$$Aw = \frac{|w|2}{\delta_\epsilon} A\left(\frac{w}{|w|} \frac{\delta_\epsilon}{2}\right) \implies |Aw| \leq \frac{2|w|}{\delta_\epsilon}\epsilon \tag{7}$$

So

$$\|A\| \leq \frac{2\epsilon}{\delta_\epsilon} \tag{8}$$

So basically, boundedness is equivalent to continuity, which is why most analysts look at bounded operators. Let's review some notation from linear algebra, given $A \in L(X, Y)$. Then,

1. $\ker(A) \subset X$ is a closed subspace. This was obvious in finite dimensions, but in infinite dimensions, you need the boundedness.
2. $\text{Im}(A) \subset Y$ is a subspace, but it is not closed, which is not as good.
3. $\text{koker}(A) \coloneqq Y/\text{Im}(A)$ is also a vector space, but it can be nasty, since if $\text{Im}(A)$ is not closed, then the quotient space is not Hausdorff.

Definition 1.2 (Equivalent Norms)

2 norms $|\cdot|_1, |\cdot|_2$ on X are equivalent if $\exists c_1 \leq c_2$ s.t. $\forall v \in X$,

$$c_1|v_1| \leq |v|_2 \leq c_2|v_1| \tag{9}$$

It is easy to see that equivalent norms induce the same topology, and equivalence of norms is an equivalence relation.

Theorem 1.2 (All Norms on a Finite-Dimensional Space Are Equivalent)

Let X be any finite-dimensional vector space. Then, any two norms on X are equivalent.

Proof. Choose a basis $\{e_1, \dots, e_n\}$ for X. Let's define the Euclidean norm $\| \sum_{j=1}^n x_j e_j \|_2 \coloneqq \sqrt{\sum_j x_j^2}$. Let $|\cdot|$ be any other norm on X. Then, by the triangle inequality, followed by Cauchy-Schwarz,

$$\left| \sum_{j=1}^n x_j e_j \right| \sum_j |x_j||e_j| \leq \sqrt{\sum_j x_j^2} \underbrace{\sqrt{\sum_j |e_j|^2}}_{=c_1} \implies |v| \leq c_1 \|v\|_2 \tag{10}$$

Now let $S^{n-1} = \|\cdot\|^{-1}(1)$, i.e. the preimage of the Euclidean norm of 1. So $|\cdot|$ is continuous w.r.t. the $\|\cdot\|_2$-topology,[a] Since it is well known that S^{n-1} is compact, $|\cdot|$ attains its minimum on S^{n-1}, call it c_2. For $v \neq 0$,

$$|v| = \|v\|_2 \left| \frac{v}{\|v\|_2} \right| \geq c_2 \|v_2\| \tag{11}$$

We mixed up the c_1, c_2 labels, but the idea is the same.

$$c_2\|v\| + 2 \leq |v| \leq c_1 \|v\|_2 \tag{12}$$

[a] it is trivially continuous w.r.t. its own topology that it induces.

This establishes that we know the metric structure of $\mathbb{R}^n$ very well, and so $\mathbb{R}^n$ is pretty boring. Unless you're in stats and you study very high-dimensional analysis.

Corollary 1.3 (1.2.6)

Every finite dimensional normed vector space is Banach.

Proof. Every finite-dimensional vector space is isomorphic to $\mathbb{R}^n$, which is complete under all norms.

Corollary 1.4 (1.2.7)

Let X be a normed vector space. Then every finite-dimensional subspace is closed.

Proof. Let $Y \subset X$ be finite dimensional. Then Y is complete, hence is closed.

Corollary 1.5 (Heine-Borel in Finite-Dimensional Normed Spaces)

Let X be a finite-dimensional normed vector space. Then $K \subset X$ is compact iff it is closed and bounded.

Corollary 1.6 (1.2.9)

Let X, Y be two normed vector spaces, with $\dim(X) < +\infty$. Then every linear function $f : X \to Y$ is bounded.

Proof. Let $A : X \to Y$ be linear. Define the graph norm on X to be

$$|w|_A \coloneqq |w|_X + |Aw|_Y \tag{13}$$

But this is equivalent to Euclidean norm since all norms on finite-dimensional vector spaces are equivalent. So $\exists c_1$ s.t.

$$|w|_X + |Aw|_Y \leq c_1 |w|_X \implies |Aw|_Y \leq c_1 |w|_X \tag{14}$$

So it tells us that any linear map from a finite-dimensional vector space to any vector space is continuous.

Theorem 1.7 (1.2.11)

Let X be a normed vector space, $B \subset X$ denote the unit ball, $S \subset X$ denote the unit sphere, then TFAE.

1. $\dim(X) < +\infty$.
2. B is compact.
3. S is compact.

Proof. If this is Hilbert space, then the proof is trivial. But this is hard in Banach spaces. It's analogous to doing linear vs nonlinear PDEs.

2 Jan 16

You should be very careful with your intuition in an infinite-dimensional vector space. You should develop this intuition on what is or isn't obvious in an infinite-dimensional vector space, so we're going over some seemingly "obvious" results.

Definition 2.1 (Hilbert Space)

A **Hilbert space** is a complete inner product space.

In analysis, you want to solve problems by construting sequences. It's bad if your vector space is not complete, i.e. these sequences don't converge.

Lemma 2.1 (Nearest Point in a Closed Convex Subset of a Hilbert Space)

Let K be a closed, nonempty convex subset of Hilbert space H. Then, there exists a unique element of minimal norm in K.

Proof. You should remember this proof since it's used in the Riesz Representation theorem's proof. Let $\delta = \inf_{x \in K} |x|$. Pick a sequence $(x_j) \subset K$ s.t. $|x_j| \to \delta$ as $j \to +\infty$. We claim that (x_j) is Cauchy. This is somewhat nonobvious, so as a proof, let $\epsilon > 0$. Then, there exists a $N \in \mathbb{N}$ s.t. $j > N \implies |x_j|^2 \leq \delta^2 + \frac{\epsilon^2}{4}$. So by convexity,

$$m, n > N \implies \frac{x_m + x_n}{2} \in K \tag{15}$$

Since δ is an infimum,

$$\left| \frac{x_m + x_n}{2} \right| \geq \delta \implies |x_m + x_n|^2 \geq 4\delta^2 \tag{16}$$

By expanding the quadratic

$$|x_m - x_n|^2 = |x_m|^2 + |x_n|^2 - 2\langle x_m, x_n \rangle \tag{17}$$

$$= -|x_m + x_n|^2 + 2|x_m|^2 + 2|x_n|^2 \tag{18}$$

$$< -4\delta^2 + 4\delta^2 + 4\frac{\epsilon^2}{4} = \epsilon^2 \tag{19}$$

which implies that $|x_m - x_n| < \epsilon$ if $m, n > N$. So, it is Cauchy[a] and since H is complete, (x_j) has a limit—call it $x_\infty \in K$ (since K is closed) with $|x_\infty| = \delta$. Now we prove uniqueness. Suppose $y \in K$ satisfies $|y| = \delta \implies \frac{x_\infty + y}{2} \in K \implies |x_\infty + y|^2 > 4\delta^2$. So, $|x_\infty - y|^2 = 0 \implies x_\infty = y$.

[a] Check if this is related to strongly Cauchy?

Theorem 2.2 (Riesz Representation Theorem)

Let H be a Hilbert space. Define $\Lambda : H \to H^*$ by $\Lambda(x)(y) \coloneqq \langle y, x \rangle$. Then, Λ is an isometeric isomorphism.

Proof. For $x \neq 0$, we have

$$\|\Lambda(x)\| = \sup_{y \neq 0} \frac{|\Lambda(x)(y)|}{|y|} = \sup_{y \neq 0} \frac{|\langle y, x \rangle|}{|y|} \leq \frac{|x||y|}{|y|} = |x| \tag{20}$$

For the other direction,

$$\frac{|\Lambda(x)(x)|}{|x|} = \frac{|x|^2}{|x|} = |x| \tag{21}$$

Therefore, $\|\Lambda(x)(x)\| = |x|$, and Λ is an isometry.

1. *Injective.* It is easy to show that $\Lambda : H \to H^*$ is injective.
2. *Surjective.* This is more involved, and we need convexity. For simplicity, let our field be $\mathbb{R}$. We've shown that $\Lambda : H \to H^*$ is an isometric injection. Let $\phi \in H^* \setminus \{0\}$. There are 2 families of natural convex sets: affine spaces and balls. It is easy to see that $K = \phi^{-1}(1)$ is an affine space and hence convex. Let x_0 be an element of minimal norm of K. We claim that $x_0 \perp \ker(\phi)$.

Indeed, we have

$$\begin{aligned} y \in \ker(\phi) &\implies x_0 + ty \in K \text{ for all } t \in \mathbb{R} && (22)\\ &\implies |x_0|^2 \leq |x_0 + ty|^2 = |x_0|^2 + 2t\langle x_0, y\rangle + t^2|y|^2 && (23)\\ &\implies 0 \leq 2t\langle x_0, y\rangle + t^2|y|^2 && (24) \end{aligned}$$

Take $t = -\frac{\langle x_0, y\rangle}{|y|^2}$, then we have

$$0 \leq \frac{-2\langle x_0, y\rangle^2}{|y|^2} + \frac{\langle x_0, y\rangle^2}{|y|^2} \implies 0 \leq -\frac{\langle x_0, y\rangle^2}{|y|^2} \tag{25}$$

and so $\langle x_0, y\rangle = 0$. So to summarize, $\phi(x_0) = 1$ with $x_0 \perp \ker(\phi)$. Let $x \in H$. Then,

$$\phi\big(x - \phi(x)x_0\big) = \phi(x) - \phi(x)\underbrace{\phi(x_0)}_{=1} = 0 \tag{26}$$

But this means

$$\begin{aligned} x - \phi(x)x_0 \in \ker(\phi) &\implies 0 = \langle x - \phi(x)x_0, x_0\rangle && (27)\\ &\implies \langle x, x_0\rangle = \phi(x)\langle x_0, x_0\rangle = \phi(x)|x_0|^2 && (28)\\ &\implies \phi(x) = \left\langle x, \frac{x_0}{|x_0|^2}\right\rangle && (29)\\ &\implies \phi = \Lambda\left(\frac{x_0}{|x_0|^2}\right) && (30) \end{aligned}$$

and therefore, Λ is surjective.

Definition 2.2 (Orthogonal Subspace)

Let $S \subset H$ be a subspace of a Hilbert space, and let

$$S^\perp \coloneqq \{x \in H \colon \langle s, x\rangle = 0 \forall s \in S\} \tag{31}$$

Theorem 2.3 (1.4.6)

Let $E \subset H$ be a closed subspace of Hilbert space H. Then, $H = E \oplus E^\perp$.

Proof. If $x \in E \cap E^\perp$, then $x \perp x \iff \langle x, x = 0 \iff x = 0$. If $x \in H$, then $x + E$ is a closed convex set, which implies that there exists an element x_0 of least norm in $x + E$. Therefore, $x_0 \perp E$ as before (previous argument: $|x_0 + ty|^2 \geq |x_0|^2$...), which implies that

$$x = \underbrace{x - x_0}_{\in E} + \underbrace{x_0}_{\in E^\perp} \tag{32}$$

More elementary properties are in the book, but now about Banach algebras, which are infinite-dimensional analogue of matrices. Remember $\mathcal{L}(X) \coloneqq \mathcal{L}(X, X)$ as the set of bounded endomorphisms on X. Remember if Y is Banach, then $L(X, Y)$ is Banach w.r.t. the operator norm.

Definition 2.3 (Banach Algebra)

A **Banach algebra** is a Banach space A equipped with a bilinear map

$$A \times A \to A, \qquad (a, b) \to ab \tag{33}$$

called the *product*, satisfying for all $a, b, c \in A$,
1. *Associativity.* $(ab)c = a(bc)$.
2. *Respects norm.* $|ab| \leq |a||b|$.

It is called
1. **commutative** if $ab = ba$ for all $a, b \in A$.
2. **unital** if there exists a $1_A \in A$ s.t. $1_A a = a1_A = a$ for all $a \in A$.

Definition 2.4 (Invertible Element of a Banach Algebra)

An element $a \in A$ of a unital Banach algebra is called **invertible** if there exists a $b \in A$ s.t. $ab = ba = 1_A$. It is unique if it exists and is denoted a^{-1}.

Here is the basic example to hold in mind.

Theorem 2.4 (Banach Algebra of Bounded Linear Endomorphisms)

$\mathcal{L}(X)$ is a unital Banach algebra w.r.t composition as multiplication.

Proof. The identity map is the unity element. It is associate since composition is associative. So, we just need to prove the norm inequality. Define

$$|ab| \coloneqq \sup_{x \neq 0} \frac{|abx|}{|x|} = \sup_{x \neq 0} \frac{|a(bx)|}{|x|} \leq \sup_{x \neq 0} \frac{|a||bx|}{|x|} = |a| \cdot \sup_{x \neq 0} \frac{|bx|}{|x|} = |a||b| \tag{34}$$

Note that this inequality can be strict since

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{1} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{1} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{0} \tag{35}$$

or just take any invertible map A, A^{-1} where $AA^{-1} = I$.

Definition 2.5 (Compact Linear Operator)

A $f \in \mathcal{L}(X)$ (or $\mathcal{L}(X, y)$) is called **compact** if for all bounded sets $B \subset X$, $f(B)$ is precompact, i.e. $\overline{f(B)}$ is compact. So, define

$$K(X) \coloneqq \{f \in \mathcal{L}(X) \text{ s.t. } f \text{ is compact}\} \tag{36}$$

$K(X)$ is very analogous to finite-rank matrices and are the first thing to study.

Theorem 2.5 (Compact Linear Endomorphisms are Banach Algebra)

$K(X)$ is a Banach algebra, but it is nonunital since the identity is not a compact map.

The following example comes up a lot in K-theory, analysis, and most of algebraic geometry is based on this.

Example 2.1 ($C(\Omega)$ as a Commutative Unital Banach Algebra)

Let Ω be a metric space. Then, $C(\Omega)$ is a commutative unital Banach algebra w.r.t. the supremum norm.

3 Fourier Analysis

The *Fourier transform* $\mathcal{F}$ is an operator (an isometric isomorphism) mapping one function to another. There are three spaces where it is naturally defined, the *Schwartz class* $\mathcal{S}$, the L^2 space, and the Banach dual of $\mathcal{S}$.

$$\begin{aligned}\mathcal{F} : \mathcal{S} &\longrightarrow \mathcal{S} \\ \mathcal{F} : L^2 &\longrightarrow L^2 \\ \mathcal{F} : \mathcal{S}' &\longrightarrow \mathcal{S}'\end{aligned}$$

and $\mathcal{S} \subset L^p \subset \mathcal{S}'$. $\mathcal{S}$ is small, $\mathcal{S}'$ is big, and L^p spaces live in between them. The *Schwartz class* is the function space of all functions $f : \mathbb{R}^n \longrightarrow \mathbb{C}$ satisfying

1. $f \in C^\infty(\mathbb{R}^n, \mathbb{C})$
2. f decays rapidly in the sense that for any scalar α, f multiplied by any monomial x^α is bounded everywhere:
$$\sup_{x \in \mathbb{R}^n} |x^\alpha f(x)| < \infty, \text{ or a similar condition } \|x^\alpha f\|_\infty < \infty \tag{37}$$
Furthermore, with vector $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{N}^n$ and multi-indexing $D^\beta = D_1^{\beta_1} D_2^{\beta_2} \dots D_n^{\beta_n} f$, all of its derivatives are also bounded for all β
$$\sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty, \text{ or } \|x^\alpha D^\beta f\|_\infty < \infty \tag{38}$$

Now Schwartz functions have both the properties of smoothness and decay. The purpose of the Fourier transform is to convert between the properties of smoothness and decay. That is, if f decays (is smooth) to some extent, then $\mathcal{F}f$ is smooth (decays) to that extent. $\mathcal{F} : \mathcal{S} \longrightarrow \mathcal{S}$ is defined (note that x and ξ are just dummy variables)

$$f(x) \mapsto \mathcal{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{-ix\xi}\, dx \tag{39}$$

This transform is a simple map from a larger class called the *Fourier integral operators*, which are maps of the form

$$f(x) \mapsto \mathcal{F}f(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{\varphi(x,\xi)} a(\xi, x)\, dx \tag{40}$$

Now since $\mathcal{F}$ is an isometric automorphism of $\mathcal{S}$, we can take its inverse transform, defined

$$\mathcal{F}^{-1} g(x) = \int_{\mathbb{R}^n} g(\xi)\, e^{ix\xi}\, d\xi \tag{41}$$

The Fourier transform allows us to connect multiplication and differentiation. Let us calculate the α-fold derivative along ξ of the Fourier transform of f.

$$\begin{aligned}\partial_\xi^\alpha \hat{f}(\xi) &= \partial_\xi^\alpha \int_{\mathbb{R}^n} e^{-ix\xi} f(x)\, dx \\ &= \int_{\mathbb{R}^n} \partial_\xi^\alpha e^{-ix\xi} f(x)\, dx \\ &= \int_{\mathbb{R}^n} (-ix)^\alpha e^{-ix\xi} f(x) \\ &= \mathcal{F}((-ix)^\alpha f(x))\end{aligned}$$

So, a α-fold differentiation after Fourier transforming a function is the same as multiplying x^α first and then Fourier transforming the same function. Now, if we multiply a the Fourier transform of a function by a

monomial this sort of looks like the Fourier transform of the derivative of the function.

$$\begin{aligned}\xi^\alpha \hat{f}(\xi) &= \int_{\mathbb{R}^n} \xi^\alpha e^{-ix\xi}\, f(x)\, dx \\ &= \int_{\mathbb{R}^n} \partial_x^\alpha (e^{-ix\xi})\, f(x)\, dx \\ &= \int_{\mathbb{R}^n} (e^{-ix\xi})\, \partial_x^\alpha f(x)\, dx \qquad \text{(Integration by Parts)}\end{aligned}$$

Minkowski Inequality says that if you have an $L^p(\Omega;\mathbb{R})$ space with norm $\|\cdot\|_{L^p} \coloneqq (\int |f|^p)^{1/p}$, then the norm satisfies the triangle inequality. Using induction gives

$$\left|\left| \sum_j f_j(x) \right|\right|_{L^p(dx)} \leq \sum_j \left|\left| f_j(x) \right|\right|_{L^p(dx)} \tag{42}$$

That is, the norm of the sum of the functions is less than the sum of the function norms. But if we index $f_j(x)$ as $f(x,y)$, then since integration is really the same thing as summation, we can write the above as

$$\left|\left| \int_{\mathbb{N}} f(x,y)\, dy \right|\right|_{L^p(dx)} \leq \int_{\mathbb{N}} \left|\left| f(x,y) \right|\right|_{L^p(dx)} dy \tag{43}$$

expanding

$$\left(\int_\Omega \left| \int_{\mathbb{N}} f(x,y)\, dy \right|^p dx \right)^{\frac{1}{p}} \leq \int_{\mathbb{N}} \left(\int_\Omega \left| f(x,y) \right|^p dx \right)^{\frac{1}{p}} dy \tag{44}$$

So summing $\int dy$, then power to p, then norming $\int dx$, then power to $\frac{1}{p}$ is less than powering to p first, then norming $\int dx$, then power to $\frac{1}{p}$, and finally summing $\int dy$. Minkowski inequality is especially useful for functions of the form

$$f(x,y) = \phi_\epsilon(x-y)\, u(y) \tag{45}$$

where ϕ_ϵ is some sort of bump function ($\phi_\epsilon(x) \to \delta(x)$ as $\epsilon \to 0$. Then, defining the convolution operator

$$(J_\epsilon u)(x) = \int \varphi_\epsilon(x-y) u(y)\, dy \tag{46}$$

Now we can approximate the delta function $\delta(x-y)$ with C^∞ bump functions by "squeezing them" as $\phi_\epsilon(t) = \phi(t/\epsilon)\epsilon^{-n}$ (we power to n to normalize over $t \in \mathbb{R}^n$, an n-dimensional input). Therefore, we can approximate u with arbitrary accuracy with a C^∞ function.

$$\underbrace{(J_\epsilon u)(x) = \int \varphi_\epsilon(x-y) u(y)\, dy}_{\text{smooth}} \approx \int \delta(x-y) u(y)\, dy = u(x) \tag{47}$$

Assuming that some smooth and compactly supported $\varphi \in C_C^\infty(\mathbb{R}^n)$ exists (we can construct this), we can construct a family of approximate identities/deltas (since deltas are identities for convolutions), called *mollifiers* (which "mollify" or smooth out an arbitrary function)

$$\{\varphi_\epsilon\}_\epsilon \text{ s.t. } \varphi_\epsilon(t) = \epsilon^{-n} \varphi(t/\epsilon) \tag{48}$$

We have

$$(J_\epsilon u)(x) \coloneqq (\varphi_\epsilon * u)(x) = \int \varphi_\epsilon(x-y) u(y)\, dy \to u(x) \text{ in } L^p \tag{49}$$

So, $\|(J_\epsilon u(x) - u(x)\|_{L^p} \to 0$ using Minkowski inequality. This property is very useful because if we're getting an arbitrary signal, we can interpret it as a function, assume the white noise is Gaussian, and then the convolution would be a smooth signal. So in reality, we don't have to worry about smoothness.

Theorem 3.1 (Fundamental Lemma of the Calculus of Variations)

If there exists some function u such that

$$\int u\varphi = 0 \text{ for all } \phi \in C_C^\infty \tag{50}$$

Then $u = 0$ almost everywhere.